Latest version: https://vuphan314.github.io
This version: 2021/10/04

Vu Hoang Nguyen Phan

Curriculum Vitae

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Vu Phan is a Computer Science Ph.D. student of Prof. Moshe Vardi at Rice University, expecting to graduate by 2022 August. He solves stochastic satisfiability with dynamic programming and knowledge compilation.
(Vu Phan is a citizen of Vietnam. He was issued an F1 visa and is currently staying in Texas.)

Education

2022/08 **Doctor of Philosophy**, *Computer Science*, Rice University, 3.61/4.00.

2019/12 **Master of Science**, *Computer Science*, Rice University, 3.60/4.00.

2017/08 **Bachelor of Science**, *Computer Science and Math*, Texas Tech University, 4.00/4.00.

Experience

2021 Summer **Graduate Technical Intern**, *Intel Corporation*, Formal Verification.

- Wrote context-free grammar to convert SystemVerilog scoreboards from one formal tool to another.
- Developed scripts to run experiments comparing two formal tools and report results.

Certification

2021 **SVA, Formal, and JasperGold Fundamentals for Designers v19.03**, *Cadence Design Systems*.

Programming Languages

Hardware **SystemVerilog**.

Software **Python, C++; Dafny, Coq**.

Public Projects

2020-2021 **ProCount**, *Projected Counter*, `https://github.com/vardigroup/DPMC`.

- Implemented a weighted projected #SAT framework with Jeffrey Dudek.

2019-2020 **DPMC**, *Dynamic-Programming Model Counter*, `https://github.com/vardigroup/DPMC`.

- Implemented a weighted #SAT framework with Jeffrey Dudek.

2019-2020 **ADDMC**, *Algebraic-Decision-Diagram Model Counter*, `https://github.com/vardigroup/ADDMC`.

- Implemented a weighted #SAT solver.

2016-2018 **LED**, *Language of Effective Definitions*, `https://vuphan314.github.io/LED`.

- Translated the literate programming language `LED` into `SequenceL` and `LaTeX`.

2015-2016 **L**, *Logic*, `https://github.com/iensen/LtoASPtranslator`.

- Translated the logic programming language `L` into `Answer Set Prolog` with Evgenii Balai.

University Courses

2017-2021 **Graduate**.

- Reasoning about Software
- Bioinformatics: Sequence Analysis
- Statistical Machine Learning
- Computer Systems Architecture
- Multi-Core Computing
- Compiler Construction
- Artificial Intelligence
- Programming Languages
- Logic in Computer Science

2014-2017 **Undergraduate**.

- Operating Systems
- Database Systems
- Computer Architecture
- Programming Languages
- Software Engineering
- Algorithms
- Object-Oriented Programming
- Computer Organization and Assembly Language Programming
- Modern Digital System Design
- Data Structures
- Automata
- Programming Principles

Bibliography

Vu Phan is the corresponding author of the following publications (authors sorted by surnames).

Thesis

[Pha19] Vu H. N. Phan. "Weighted Model Counting with Algebraic Decision Diagrams". MS thesis. Rice University, 2019. URL: `https://scholarship.rice.edu/handle/1911/107761`.

Conference Papers

[DPV21] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "ProCount: Weighted Projected Model Counting with Graded Project-Join Trees". In: *Conference on Theory and Applications of Satisfiability Testing (SAT)*. 2021. URL: `http://jeffreydudek.com/papers/sat21_procount.pdf`.

[DPV20a] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "DPMC: Weighted Model Counting by Dynamic Programming on Project-Join Trees". In: *Conference on Principles and Practice of Constraint Programming (CP)*. 2020. URL: `https://arxiv.org/abs/2008.08748`.

[DPV20b] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "ADDMC: Weighted Model Counting with Algebraic Decision Diagrams". In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2020. URL: `https://arxiv.org/abs/1907.05000`.

Workshop Paper

[Pha18] Vu H. N. Phan. "Syntactic Conditions for Antichain Property in Consistency Restoring Prolog". In: *Workshop on Answer Set Programming and Other Computing Paradigms (ASPOCP)*. 2018. URL: `https://arxiv.org/abs/1809.09319`.